



TILERA[®]

***TILENCORE-GX
INTELLIGENT
APPLICATION ADAPTER
USER'S GUIDE***

Featuring the TILE-Gx16 or TILE-Gx9 Processor

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TILERA CORPORATION

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Contact Information:

Tiler Corporation
Information info@tilera.com
Web Site http://www.tilera.com

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CHAPTER 1 OVERVIEW

1.1 About This Manual

Thank you for purchasing the TILEncore-Gx16™ or TILEncore-Gx9™ Intelligent Application Adapter featuring Tilera's revolutionary multicore Tile Processor™. This manual contains information for setting up and evaluating the TILEncore-Gx card.

1.2 In this Release

The following items are included in this shipment:

- TILEncore-Gx16, featuring the TILE-Gx16 processor. The board comes installed with the TILE-Gx8016™ (Series 8000) processor.
- OR—
- TILEncore-Gx9, featuring the TILE-Gx9 processor. The board comes installed with the TILE-Gx8009™ (Series 8000) processor.
- Standard low-profile or Half Height I/O Bracket.
- Standard mini-to-A USB cable.

1.3 About this Guide

This Guide describes the installation, operating requirements, setup of the TILEncore-Gx and features of Tilera's TILE-Gx16 or TILE-Gx9 processor. This book includes the following chapters:

- [Chapter 1: Overview](#) (*this chapter*) describes the contents of this manual.
- [Chapter 2: About the TILEncore-Gx16 or TILEncore-Gx9](#) identifies the intended user for the TILEncore-Gx and the board's features and supported interfaces.
- [Chapter 3: System and Board Requirements](#) specifies the power and system requirements, including the hardware and software requirements that should be followed in order to operate the board properly.
- [Chapter 4: Installing the TILEncore-Gx16 or TILEncore-Gx9](#) provides pre-installation guidelines and proper handling instructions.
- [Chapter 5: Functional Description](#) describes some of the board's features and subsystems, specifically the board resets, LEDs, interrupts, and boot options.
- [Chapter 6: Running Programs on the TILEncore-Gx16 or TILEncore-Gx9](#) provides instructions for booting the board and running a sample program.
- [Appendix A: Specifications](#) provides electrical, mechanical, and environmental specifications for the board.
- [Appendix B: Safety and Regulatory Compliance](#) lists the safety and compliance information for the TILEncore-Gx.

- [Appendix C: Troubleshooting](#)
- [Glossary](#) provides definitions for terms used in this document.

An [Index](#) is also included.

1.3.1 Intended Audience

This book is intended primarily for hardware developers who want to use the TILEncore-Gx to jump start whole system designs, designers evaluating the TILE-Gx™ processor for use in their boards, and individuals looking for a system solution. The TILEncore-Gx is offered separately from but in conjunction with the Multicore Development Environment™ (MDE), accompanying tools, software libraries, and software documentation. Contact your support engineer or sales representative for more information about the MDE.

1.3.2 Notation Conventions

[Table 1-1](#) lists document notation conventions.

Table 1-1. Notation Conventions

Example	Description
Close command (File menu)	Titles in reference sections indicate the location of an item within the Graphical User Interface (GUI) for the MDE (for example, the Close command appears on the File menu).
Write In Progress (WIP) bit	Courier text indicates the names of: • A bit or bitfield, for example the CHANNEL bit • A special purpose register (SPR), for example the RSH_COORD SPR • Code sample • Application, for example the tile-monitor application • Command, for example the link command • State, for example the IDLE state
RSHIM MMIO registers	Blue and underlined text in Courier font indicates these are hypertext links to HTML files associated with the interface, or links to a specific web site.
Appendix B: Safety and Regulatory Compliance	Blue text in text font (Palatino) indicates a (cross-reference) Hypertext link to text elsewhere in the manual. If you are reading a soft-copy of this manual, you can click on the link to jump directly to the referenced section.
{this that}	Alternative required items in syntax descriptions appear within curly brackets and separated by vertical bars; read the example as this or that. One or the other is required.
[this that]	Optional items in syntax descriptions appear within brackets and separated by vertical bars; read the example as an optional this or that.
[this,...]	Optional item lists in syntax descriptions appear within brackets delimited by commas and terminated with an ellipse; read the example as an optional comma-separated list of this.
.SECTION	Commands, directives, keywords, and feature names are in text with Arial font.
<filename>	Non-keyword placeholders appear in text in italic style format and within brackets.
www.tilera.com	Worldwide web addresses, directory names, code samples all share a courier font.

Table 1-1. Notation Conventions (continued)

Example	Description
	Note: For correct operation, ... A Note provides supplementary information on a related topic. In the online version of this book, the word Note appears instead of this symbol.
	Caution: Incorrect device operation might result if ... Caution: Device damage might result if ... A Caution identifies conditions or inappropriate usage of the product that could lead to undesirable results or product damage. In the online version of this book, the word Caution appears instead of this symbol.

1.4 Technical and Customer Support

This section describes how to obtain additional information.

1.4.1 Customer Support

You can reach Tiler Corporation Customer Support in either of two ways:

- Visit the Tiler Corporation Web site at
<http://www.tiler.com/>
- E-mail questions to
support@tiler.com

1.4.2 Product Information

You can obtain additional product information from the Tiler Corporation Web site, from the product CD-ROM, or from Tiler printed publications.

Access the Tiler Corporation Web site online at:

<http://www.tiler.com/>

1.4.3 Related Documents

Additional documentation is available upon request. Most of the documentation listed below is included with Tiler's MDE.

1.4.3.1 Hardware Documents

Hardware Design

- *TILE-Gx16 Processor Data Sheet (DS401)*
—OR—
- *TILE-Gx9 Processor Data Sheet (DS403)*
- *Hardware Design Guidelines for TILE-Gx36, TILE-Gx16, and TILE-Gx9 Processors (WP006)*

Architecture

- *Architectural Overview for the TILE-Gx Series (UG130)*
- *TILE-Gx Instruction Set Architecture (UG401)*
- *Tile Processor and I/O Device Guide for the TILE-Gx Family of Processors (UG404)*

Boards

- *TILEncore-Gx Intelligent Application Adapter User's Guide Featuring the TILE-Gx16 or TILE-Gx9 Processor (this manual)*
- *Installation Sheet*
 - *For TILEncore-Gx16 (IS108)*
- OR—
- *For TILEncore-Gx9 (IS109)*
- *Tilera Board Test Kit Guide for TILE-Gx Processors (UG422)*

1.4.3.2 Software Documents

The Tilera MDE Document Index provides links to all the MDE documentation, including HTML and PDF files and additional documentation:

```
${TILERERA_ROOT}/doc/index.html
```

Note: `${TILERERA_ROOT}` is the root of the Tilera MDE install directory.

1.5 MDE Documentation

Note: `TILERERA_ROOT` is an environment variable that points to the root of the MDE installation directory. As such it is a convenient shorthand in the documentation. It is required by many Tilera tools and examples. We recommend that you use the `tile-env` script described in *Getting Started with the Multicore Development Environment (UG504)* to set it correctly.

Tilera MDE Document Index

The Tilera MDE Document Index provides links to online copies of the MDE documentation, in both HTML and PDF formats:

```
$TILERERA_ROOT/doc/index.html
```

Location of PDF Files

The PDF versions of MDE documents can be found at this location:

```
$TILERERA_ROOT/doc/pdf/
```

Man Pages and Info Pages

To view a man page for an x86 tool, run:

```
$ tile-man toolname
```

To view a man page for a Tile tool, run this on the x86 host:

```
$ man toolname
```

IDE Online Help

The IDE has its own [Tilera IDE Online Documentation](#), which you can access when you use the IDE.

To get to the online help:

- Start `tile-eclipse`.

Select the menu command **Help → Help Contents**.

- When the online help browser is displayed, select the document **Tilera IDE Online Documentation** in the left navigation pane, if it is not already selected.

CHAPTER 2 ABOUT THE TILENCORE-GX16 OR TILENCORE-GX9

2.1 Product Overview

The TILEncore-Gx16™ or TILEncore-Gx9™ adapter card is a PCIe endpoint that enables customers to develop and test software applications on the TILE-Gx16™ or TILE-Gx9™ processor. Physically, the board is a full-height, low-profile, ½-length, single-slot PCI Express form factor with an 8-lane edge connector. Up to two optical network interfaces are provided on the rear panel. TILEncore Card is a modular board with a processor and SODIMM memory.

2.2 Purpose

The TILEncore-Gx card serves three purposes: as an early development vehicle, as evaluation hardware for the multicore processor, and for integration into a “black box” system for developers who do not want to build their own hardware. Due to the wide variety of usage models, no software applications accompany the TILEncore-Gx, but they can be obtained separately (see [“Customer Support” on page 3](#)).

2.3 TILEncore-Gx Design Features

- TILE-Gx processor capable of running at 1.0 GHz.
- Single modular DDR3 (Low profile SODIMM) memory interfaces.
- Standard USB accessed via Mini-to-A cable.
- PCI Express standard, gold-fingered edge connector to host, which uses an eight-lane physical connector.
- User-selectable boot options, including SPI flash ROM and host boot via PCIe.
- The on-board serial Boot ROM(s) shipped with the prebooter (to configure the PCIe link). Initial booting is expected to be via the PCIe.
- Temperature monitoring of the environment and the TILE-Gx™ core.
- Power monitoring via software.

2.3.1 Design and Layout

The TILEncore-Gx is comprised of two primary sections:

- The TILE-Gx processor (and its memory subsystems)
- The interface subsystem with two SPF+ 10GbE ports

Figure 2-1 depicts the block diagram and Figure 2-2 shows the major card components and associations for TILEncore-Gx.

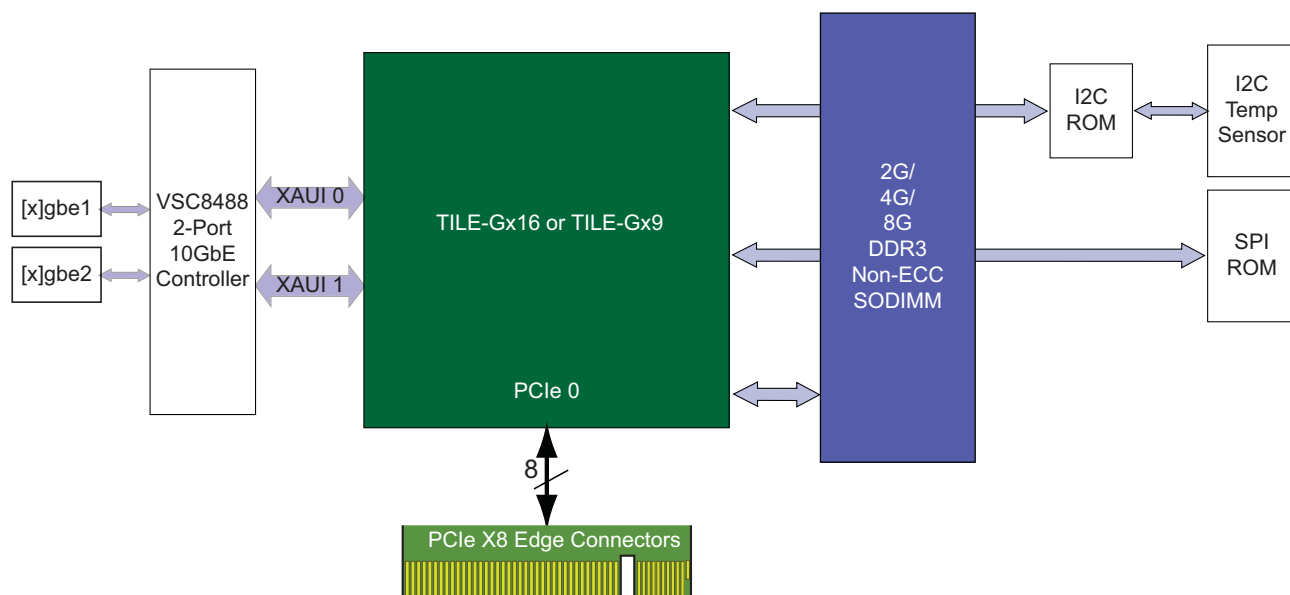


Figure 2-1: TILEncore-Gx Block Diagram

Figure 2-3 shows a top view of the board and identifies the major components. For locations of the LEDs, refer to Figure 7-1.

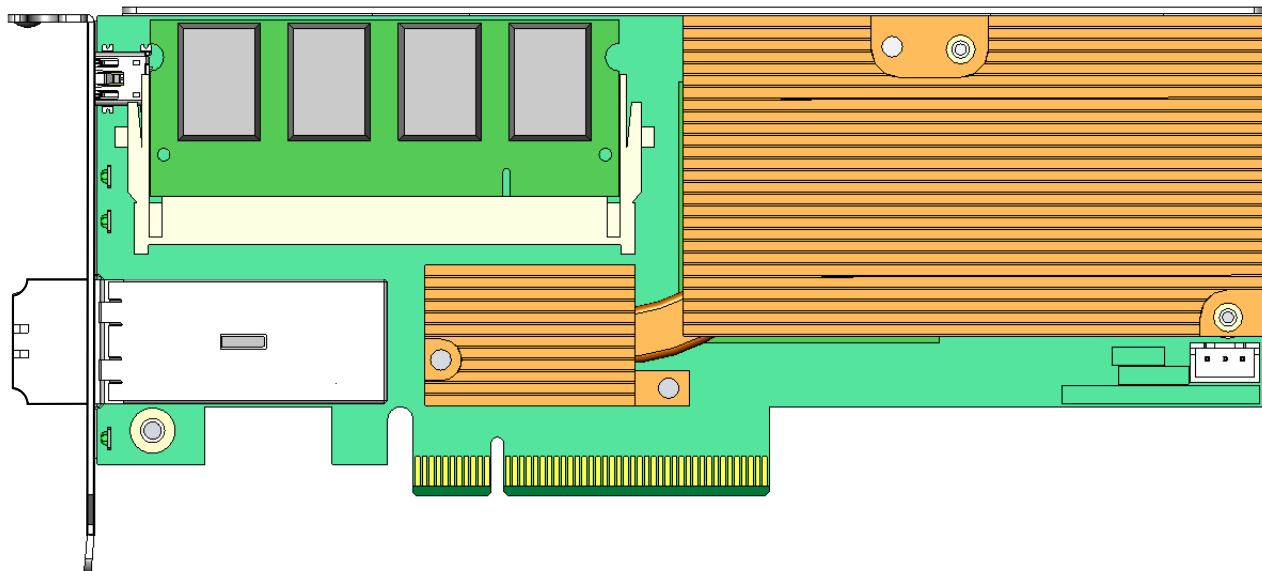


Figure 2-2: TILEncore-Gx16/Gx9 Conceptual Layout

Figure 2-3 shows the front view of the main board and the location of the jumpers.

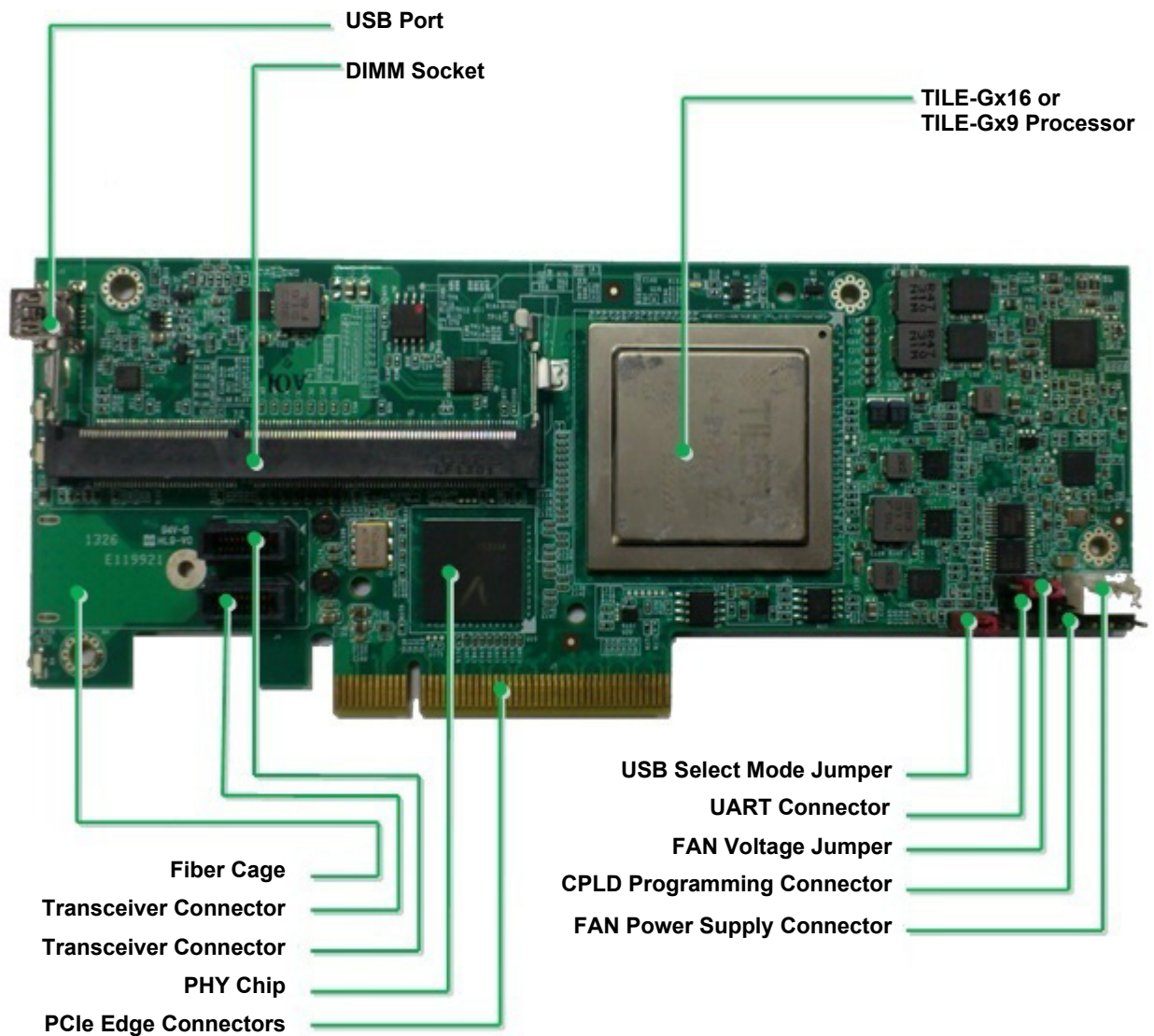


Figure 2-3: TILEncore-Gx16/Gx9 Physical Layout¹

1. Note that the Fiber Cage has been removed to show the connectors.

CHAPTER 3 SYSTEM AND BOARD REQUIREMENTS

3.1 System Requirements

This section lists minimal requirements for installing and using the TILEncore Card™.

3.1.1 Hardware Requirements

Minimum system hardware requirements are described in the sections that follow.

3.1.1.1 Host System

Tilera® has verified that the TILEncore-Gx card works on many systems. For a complete list of qualified systems, refer to “PCIe Host Requirements” in the *MDE Patch Notes* (UG1118) and *MDE Release Notes* (UG1108).

3.1.1.2 Mechanical Requirements

As a Low profile, Half-Length or Half Height, Half-Length, PCIe add-in card, the TILEncore-Gx can be installed in a standard x8 or x16 PCIe slot and conforms to PCIe clearance specifications for top and bottom component heights as well as specified keepouts.

3.1.1.3 Mother Board Requirements

The TILEncore-Gx supports a x8 PCIe interface and requires a x8 PCIe connector. Higher bandwidth connectors are backward compatible; therefore, a card can also be installed in an x16 slot. From an electrical standpoint, the TILEncore-Gx trains up to Gen 2 x8 (maximum).

3.1.1.4 Air Flow

For boards installed in host systems, adhere to the recommended operating conditions as described in the *TILEncore-Gx16 Installation Sheet* (IS108) or *TILEncore-Gx9 Installation Sheet* (IS109).

Note: Do not leave the PC cover open as it can disrupt air flow inside the PC. Air circulation is best with the cover closed.

3.1.2 Software Requirements

Minimum system software requirements are described in the sections that follow.

3.1.2.1 Operating System Requirements

In order for you to operate the TILEncore-Gx, you need the following software installed on your system:

- Operating System software should be Multicore Development Environment™ (MDE) Release 4.3 or later. Refer to the *MDE Release Notes* (UG1108) for specific requirements.

- Red Hat Enterprise Linux software. The MDE has been qualified on a host system running the appropriate releases of RHEL and CentOS, as specified in the *MDE Release Notes* (UG1108) and/or *MDE Patch Notes* (UG1118).
- Tileria includes pre-built drivers for Red Hat Enterprise Linux 6, CentOS 6, Red Hat Enterprise Linux 5, and CentOS 5. The driver source is provided; other kernels need to compile their own drivers. (Tileria provides build scripts, for example: `$TILERIA_ROOT/lib/modules/tilepci-compile`). Refer to *MDE Release Notes* (UG1108) and/or *MDE Patch Notes* (UG1118) for a list of these pre-built drivers.

The *Linux Red Hat Installation Guide* can be found at:

https://access.redhat.com/site/documentation/Red_Hat_Enterprise_Linux/

For instructions on configuring drivers, refer to “I/O Devices and Drivers” in the *Tileria MDE System Programmer’s Guide*. For more information, contact support@tilera.com.

3.1.2.2 Development Software Requirements

The following software is required:

Tileria Multicore Development Environment (MDE). You must use a current copy of Tileria’s MDE— at a minimum 4.2 revision level. This developer’s kit provides drivers and runtime code required to use the TILEncore-Gx16/Gx9.

3.2 Power Requirements

The TILEncore-Gx gets power from the PCIe edge connectors.

- 12V @ 1.6 A
- 5V @ 1.4 A

CHAPTER 4 INSTALLING THE TILENCORE-GX16 OR TILENCORE-GX9

4.1 Overview

This section describes how to install the TILEncore-Gx™ card.

4.2 Pre-Installation Guidelines

This section describes steps to follow when installing the TILEncore-Gx card.

4.2.1 Unpacking the TILEncore-Gx Card

Before unpacking your TILEncore-Gx, inspect the shipping carton for damage. If the shipping container appears damaged, notify the shipper and Tilera® immediately.

■ When you unpack your TILEncore-Gx card:

Carefully remove packing material. Retain the original packing in case the unit must be returned for servicing or shipping to another location.

4.2.2 Handling the TILEncore-Gx Card

4.2.2.1 Preventing Electrostatic Discharge (ESD)

In order to prevent ESD during shipment and storage, the TILEncore-Gx is placed in a static shielding bag, seated in an anti-static foam cushion, and placed in a box. This protection prevents vibration between the card and the box and avoids possible buildup of static electricity and ESD damage.

Before opening the box place it on an ESD-protected work area, preferably on a static dissipative table top. Workers should further protect the card by wearing a wrist strap that is connected to a ground point. The wrist strap is the primary means of controlling static charges on a person. A non-conductive flooring is also recommended as walking across a conductive surface causes static electricity to build up. When the card is not inserted in a PC, it should be kept in the protective bag in which it was shipped.

4.2.2.2 Preventing Damage to the TILEncore-Gx Card

When handling the TILEncore-Gx always hold it by the edges, not by any components, connectors or fixtures installed on the card. Do not flex the card as this can damage the solder connections.

4.2.3 Preparing for Installation or Removal

[Table 4-1](#) lists the tools needed for removing and installing the TILEncore-Gx card in your PC.

Table 4-1. Removal and Replacement Tools

Task	Tools
Removing/installing the TILEncore-Gx	• Phillips screwdriver (no. 2), depending on PCI bracket style • Antistatic (ESD) wrist strap

Note: It is important to use the correct size screwdriver in order to avoid damaging the screw heads.

4.3 TILEncore-Gx Card Installation

This section provides installation and removal guidelines only.

Note: Some motherboards have restricted slot usage. Refer to your motherboard documentation for this information.

4.3.1 Preliminary Notification

Before removing or installing any cards, follow these steps:

1. For a standard card installation, first shut down the host system. Ensure that all power is disconnected from the PC in which it is installed.
2. Place an antistatic wrist guard strap on your wrist and attach the other end of the strap to a ground point.
3. Read the instructions in [“Safety and Regulatory Compliance” on page 39](#) that describe how to handle the TILEncore-Gx card.

4.3.2 Installing a TILEncore-Gx Card

Follow the steps described below to install the hardware and software:

1. Ensure that the jumpers are set, as indicated in [Section 4.5 “Jumper Settings”](#) on page 17.
2. Power down the host system.
3. Firmly insert the TILEncore-Gx into the PCIe x8/x16 slot.
4. Fasten the card appropriately for your system, either by snapping the fastening lever or installing the PCI bracket screw securely.
5. Optional: Connect the mini-to-A USB cable, as shown in [Figure 4-2](#). Connect the other end to the host PC in an available USB port.

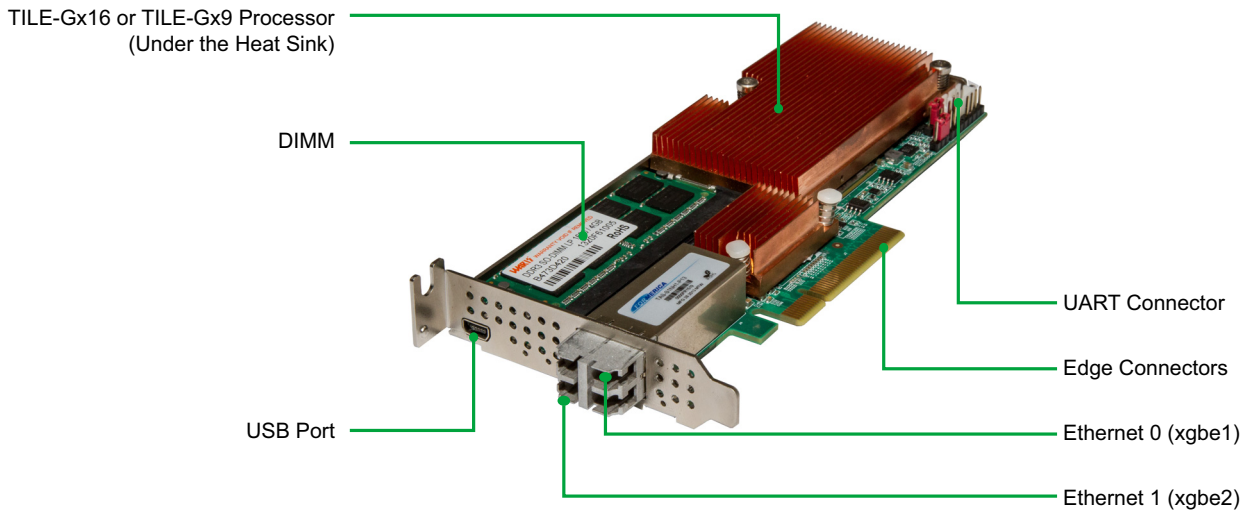


Figure 4-1: TILEncore-Gx — Front View

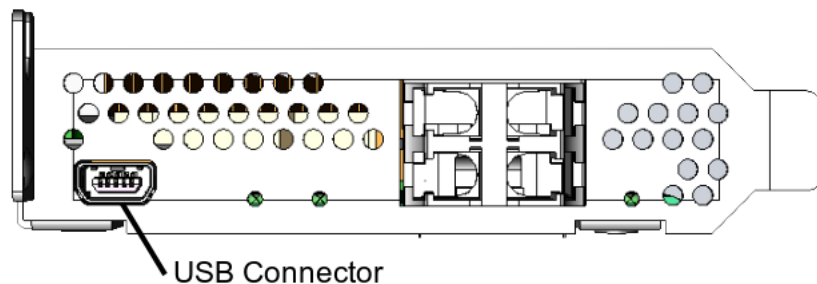


Figure 4-2: Bracket showing the USB Connector

6. Install the latest MDE release, as described in *Getting Started with the Multicore Development Environment* (UG504), or the MDE Release Notes (UG1108) supplied with the MDE. After you have completed the MDE installation process, the PDF versions of MDE documents can be found in: `$TILER_ROOT/doc/pdf/`.
7. Build the host PCIe driver by running the `$TILER_ROOT/lib/modules/tilepci-compile` script to build the x86 PCIe driver for your OS version. Refer to “Building and Installing the x86 Drivers” as described in the *Getting Started with the Multicore Development Environment* (UG504).

4.3.3 Removing a TILEncore-Gx Card

1. Power down the host system.
2. Remove the AC cord from the host system.
3. Remove the PC's cover and place it aside. Begin by deactivating the slot in which the TILEncore-Gx is installed. (Refer to the documentation provided with the PC chassis and/or the operating system.) Note that the green “RUN” LED on the TILEncore-Gx goes out, indicating that the board has been powered down and can be safely removed. Refer to [Section 5.8 “LEDs”](#) on page 29.

4. Disconnect the mini-to-A USB cable, if one is installed.
5. Remove any fiber cables prior to removing the card.
6. Loosen screws or release the lever holding the card in place. Depending on the PC you are using, the card is held in place by either a tool-less fastening lever or a screw. You might need to use a Phillips screwdriver (no. 2) to loosen the screw securing the top of the TILEncore-Gx card's bracket.
7. Remove the card from the host system and place it in an antistatic bag.

4.4 Connecting Cables

4.4.1 Mini-to-A USB Cable

Tilera provides one cable: a mini-to-A USB cable.

1. Connect the USB A connection end to the host system.
2. Connect the USB mini end of the cable to the USB port on the TILEncore-Gx card, as shown in [Figure 4-2](#).

4.4.2 Ethernet Cables

Note that Tilera does not provide cables.

1. Connect the appropriate optical (Ethernet) cables to the installed transceivers on the TILEncore-Gx.
2. Connect the other end to either a router/switch/appliance or other network equipment.

Note: The Ethernet cable for 10G interfaces is preferred since the transceiver is a 10G. While a 1G fiber cable can be used, its speed is limited to 1G only.

The following cable types might be required:

Table 4-2. Cable Types

Connection	Cables
10G Fiber with SR	10GBASE-SR OM3 Multimode Fiber (typically orange)

Specification for Ethernet connector SFF+ SR transceivers:

Table 4-2. Specification for Ethernet Connector SFF+ Transceivers

Part Number	Vendor	Description
B6110700	Formerica	(GP). DIP Optical Transceiver. Mini Dual SFF+ SR Transceiver with Cage. 10G 300m 3.3V 0-70C AC/TTL 850nm VCSEL 48.7x14.6x10.85mm. TAS-S1NH1-P13.FORMERICA

Note: The onboard connector has a fixed SFF_SR transceiver on it. You can not change the transceiver.

4.5 Jumper Settings

This section lists TILEncore-Gx jumpers with their default settings.

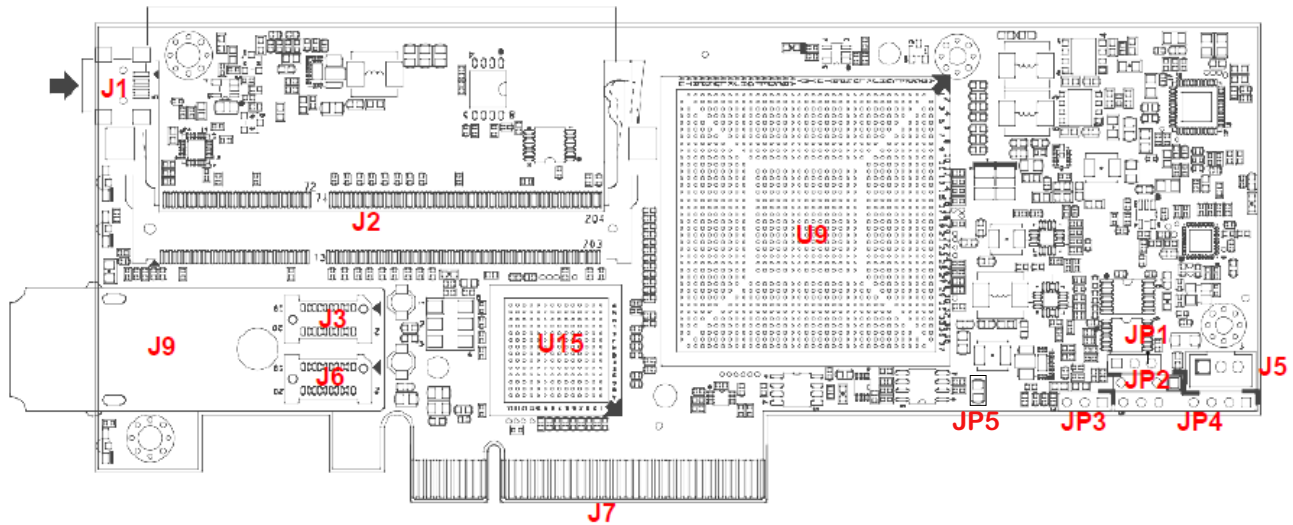


Figure 4-3: Jumper Locations

Table 4-3 lists the TILEncore-Gx jumpers and connectors settings. Default values are displayed in boldface type.

Table 4-3. Connector Definitions

Reference Designator	Jumper ID	Descriptions
J1	USB Connector	
J2	DIMM Sockets	
J3	Transceiver Connector	Ethernet 1 (xgbe2)
J6	Transceiver Connector	Ethernet 0 (xgbe1)
J5	FAN Power Supply Connector	Reserved Note that the TILEncore Card includes a passive heat sink.
J7	PCIe Bus Connectors	
J9	SFF Cage	
U9	Tilera Processor	- GX8016 —OR— - GX8009
U15	PHY Chip	

Table 4-3. Connector Definitions (continued)

Reference Designator	Jumper ID	Descriptions
JP1	FAN Voltage Jumper	Reserved
JP2	UART Connector	• Pin 1: TX • Pin 2: RX • Pin 3: NC • Pin 4: GND
JP3	USB Select Mode Jumper	• Pin 1-2: Host Mode • Pin 2-3: Endpoint Mode (default)
JP4	Programming Connector	• Pin 1: 3 V • Pin 2: CPLD TDO • Pin 3: CPLD TDI • Pin 4: BSCN/GND • Pin 5: NC • Pin 6: CPLD TMS • Pin 7: GND • Pin 8: CPLD TCK
JP5	Onboard SPI ROM Operations	Enables/disables write operations to the onboard SPI ROM. • Pin 1-2 (Open): Disable Write Protect (default) • Pin 1-2 (Close): Enable Write Protect

Table 4-4. Description on Connectors and Jumpers

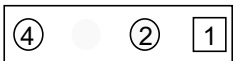
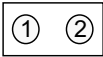
Connector	Function	Description
JP1	Fan voltage select  JP1	Fan settings for this connector are not applicable and are reserved. • Pin 1-2: 12V • Pin 2-3: 5V (default)
JP3	USB configuration  JP3	• Pin 1-2: Host Mode • Pin 2-3: Endpoint Mode (default)
JP2	UART pin header (Console port)  JP2	• Pin 1: TX • Pin 2: RX • Pin 3: NC • Pin 4: GND

Table 4-4. Description on Connectors and Jumpers (continued)

Connector	Function	Description
JP4	Programs the CPLD  JP4	• Pin 1: 3V • Pin 2: CPLD TDO • Pin 3: CPLD TDI • Pin 4: BSCN/GND • Pin 5: NC • Pin 6: CPLD TMS • Pin 7: GND • Pin 8: CPLD TCK
JP5	Enables/disables write operations to the onboard SPI ROM  JP5	• ON When the jumper is on the write operation is enabled. • OFF Disable (default).
J5	Connects to FAN device  J5	Fan settings are reserved . • Pin 1 Monitor the rate • Pin 2 Fan Voltage^a • Pin 3 PWM

a. Fan voltage is determined by JP1.

4.6 Installing TILEncore-Gx Drivers

For instructions on installing drivers, refer to “Installing this Release” in the *Getting Started with the Multicore Development Environment* (UG504).

CHAPTER 5 FUNCTIONAL DESCRIPTION

This chapter describes each of the TILEncore-Gx™ card's functional subsystems.

5.1 Power On Reset

PCIe reset is logically connected to the TILEncore-Gx from the PCIe edge connectors and is distributed from an onboard Complex Programmable Logic Device (CPLD) to the TILE-Gx™ processor. On reset it causes the TILE-Gx to execute a boot. The link automatically trains as an endpoint.

5.2 Clock Distribution

There are five clock trees on the PCIe card that are generated from a single clock driver:

Clock Speed	Description
125 MHz	This clock is distributed, single ended to the TILE-Gx processor's GX_CORE_REFCLK.
100 MHz	This is a LVPECL clock distribution for PCIe via GX_PCIE_REFCLK. Note that this clock comes from the edge connector and not the IDT5V49EE904. It is listed here because it can be used instead of the one from the IDT5V49EE904.
156.25 MHz	This clock is distributed, single ended, to the TILE-Gx processor's GX_ENET_REFCLK.
25 MHz	This clock is distributed and single ended to the CPLD.
26 MHz	This clock is distributed, single ended, to the USB3317 USB Controller USB_ULPI0_REFCLK.

5.3 TILE-Gx Processor Connections

Table 5-1 provides a list of the TILE-Gx™ processor interfaces and their connections:

Table 5-1. Tile Processor I/O Connections

Description	Interfaces to
DDR3 memory interface '0'	DDR3 SODIMM
XAUI interface '0'	VSC8488 One pair (XAUI0 and XAUI1_LAS1) are used to connect all 16 GPIOs.
XAUI interface '1'	VSC8488
PCIe interface '0'	PCIe edge connector

Table 5-1. Tile Processor I/O Connections (continued)

Description	Interfaces to
GPIO Interface	Various I/Os
UART	On-board RS-232 Transceiver
I2C Master 0	Boot/BIB ROM
I2C Master 1	DIMM SPD ROM
I2C Master 2	I ² C Peripheral devices (Temperature sensors, power monitors and clock control)
I2C Slave	N/A
SROM_SPI	On-board SPI ROM
USB0	USB Connector
USB1	Not used

5.4 I/O Subsystems

5.4.1 SR Fiber Ethernet Ports for 10G Ethernet

These connectors are located on the board and provide the 10Gps Ethernet over fiber using built-in multi-mode 10GbE optical transceivers. This XAUI connector uses port 1 and port 2. Connector details are as follows:

- Description: Connector. Slot 10Px2 P=0.8mm Vertical. SAMTEC
- Part Number: HSEC8-110-01-L-DV-A

[Table 5-2](#) and [Table 5-3](#) show the connector summary.

Table 5-2. Port 0 Connector Summary

Pin	Signal Name	Description	Type
1	GND	Receiver Ground (Common with Transmitter Ground).	Power
3	VSC8488_GPIO[1]/TXFAULT	Transmitter Fault.	Output
5	VSC8488_GPIO[11]/TXDIS	Transmitter Disable. Laser output disabled on high or open.	Output
7	SDA_SFP[1]	Definition 2. Data line for Serial ID.	Bidirectional
9	SCL_SFP[1]	Definition 1. Clock line for Serial ID.	Output
11	MOD_DET[1]	Receiver Ground (Common with Transmitter Ground).	Power
13	SFP1_RS0	No connection required.	Output
15	LOPC[1]	Loss of Signal indication. Logic 0 indicates normal operation.	Output

Table 5-2. Port 0 Connector Summary (continued)

Pin	Signal Name	Description	Type
17	SFP1_RS1	Receiver Ground (Common with Transmitter Ground).	Power
19	GND	Receiver Ground (Common with Transmitter Ground).	Power
2	GND	Receiver Ground (Common with Transmitter Ground).	Power
4	TXDATAOUT[1]-	Transmitter Inverted DATA in. AC Coupled.	Output
6	TXDATAOUT[1]+	Transmitter Non-Inverted DATA in. AC Coupled.	Output
8	GND	Receiver Ground (Common with Transmitter Ground).	Power
10	SFPP_VCCT_1	Receiver Power Supply.	Power
12	SFPP_VCCR_1	Receiver Power Supply.	Power
14	GND	Receiver Ground (Common with Transmitter Ground).	Power
16	RXDATAIN[1]+	Receiver Non-inverted DATA out. AC Coupled.	Input
18	RXDATAIN[1]-	Receiver Inverted DATA out. AC Coupled.	Input
20	GND	Receiver Ground (Common with Transmitter Ground).	Power

Table 5-3. Port 1 Connector Summary

Pin	Signal Name	Description	Type
1	GND	Receiver Ground (Common with Transmitter Ground).	Power
3	RXDATAIN[0]-	Receiver Inverted DATA out. AC Coupled.	Input
5	RXDATAIN[0]+	Receiver Non-inverted DATA out. AC Coupled.	Input
7	GND	Receiver Ground (Common with Transmitter Ground).	Power
9	SFPP_VCCR_0	Receiver Power Supply.	Power
11	SFPP_VCCT_0	Receiver Power Supply.	Power
13	GND	Receiver Ground (Common with Transmitter Ground).	Power
15	TXDATAOUT[0]+	Transmitter Non-Inverted DATA in. AC Coupled.	Output
17	TXDATAOUT[0]-	Transmitter Inverted DATA in. AC Coupled.	Output
19	GND	Receiver Ground (Common with Transmitter Ground).	Power
2	GND	Receiver Ground (Common with Transmitter Ground).	Power

Table 5-3. Port 1 Connector Summary (continued)

Pin	Signal Name	Description	Type
4	SFP0_RS1	Receiver Ground (Common with Transmitter Ground).	Power
6	LOPC[0]	Loss of Signal indication. Logic 0 indicates normal operation.	Output
8	SFP0_RS0	No connection required.	Output
10	MOD_DET[0]	Receiver Ground (Common with Transmitter Ground).	Power
12	SCL_SFP[0]	Definition 1. Clock line for Serial ID.	Output
14	SDA_SFP[0]	Definition 2. Data line for Serial ID.	Bidirectional
16	VSC8488_GPIO[10]/ TXDIS	Transmitter Disable. Laser output disabled on high or open.	Output
18	VSC8488_GPIO[0]/ TXFAULT	Transmitter Fault.	Output
20	GND	Receiver Ground (Common with Transmitter Ground).	Power

5.4.2 UART Main Board Connector

Connector Detail

- Reference designator: JP2
- Pin Header: 1Px4(-3) P=2.54mm
- Part Number: PH104S-D3-NY-L
- Vendor: TechBest

Table 5-4. RS232 Connector Summary

Pin	Serial Cable Wire Colors	Signal Name	DB-9 Female Serial RS232 Pin Number	Description
1	Black	RS232-TX	JP2 pin 1 is DB9 pin 2	TILE-Gx RS232 Data Output
2	Brown	RS232-RX	JP2 pin 2 is DB9 pin 3	TILE-Gx RS232 Data input
3	N/A	N/A	N/A	N/A
4	Red	Ground	JP2 pin 4 is DB9 pin 5	Power Return

5.5 Miscellaneous Connections

5.5.1 Fan Power Connectors

Connectors details are as follows:

- Reference designator J5
- Eci Connector
- CONN, HEADER, 3P, 2.5mm
- Eci P/N: 2532-03

Table 5-5. Fan Power Connector Summary Table.

Pin	Signal Name	Description
1	FAN_TACH	Control the rate.
2	FAN_VOLT_SW	Toggling voltage by external jumper setting that have switched +12V or +5V (default).
3	FET_FAN_GND	Ground.

5.6 DDR3 Memory Configuration Subsystem

One memory interface of the TILE-Gx is connected to a DIMM socket for easy user modification of speed and density. Though there is a target configuration of 4 Gbyte/interface density, component availability or customer requirements make these interfaces a “build option”. Configuration data for these memory interfaces is stored in the DIMM SPD (2-wire serial) ROM so that the TILE-Gx software can configure its memory controllers to work correctly with the memory.

The TILE-Gx processor has a set of four VID output pins, VID_DDR[3 : 0], that are designed to drive the VID[6 : 3] pins on the regulator. For the TILEcore-Gx16 and TILEcore-Gx9 cards, memory voltage can be configured as follows:

Table 5-6. Memory Voltage Configuration

VID Output Pin	VID Pin on the Regulator	Options
VID_DDR[1]	VID[4]	• 1 High 1.5V • 0 Low 1.35V

Note: Only Low-profile SODIMMs can be used in the TILEcore-Gx.

5.6.1 SODIMM Connectors Detail

DIMM Connector Descriptions

Connector. DDR3 SO-DIMM 204P H=9.2mm STD Black.AAA-DDR-111-K01.LOTES

SODIMM Connector Assignments

The DIMM connector numbers include SODIMM MEM0 J2.

These connector assignments follow the standard DDR3 SODIMM JEDEC specification.

5.7 TILE-Gx16 or TILE-Gx9 Processor's GPIO

Table 5-7. TILE-Gx16 or TILE-Gx9 Processor's GPIO Assignment

Pin Name	Usage Class	Usage Detail
GPIO[0]	Test Point	Test Point
GPIO[1]	Test Point	Test Point
GPIO[2]	NC	NC
GPIO[3]	NC	NC
GPIO[4]	NC	NC
GPIO[5]	NC	NC
GPIO[6]	NC	NC
GPIO[7]	NC	NC
GPIO[8]	NC	NC
GPIO[9]	NC	NC
GPIO[10]	NC	NC
GPIO[11]	NC	NC
GPIO[12]	NC	NC
GPIO[13]	NC	NC
GPIO[14]	NC	NC
GPIO[15]	NC	NC
GPIO[16]	10G_LED_Link_0	LED Link
GPIO[17]	10G_LED_Link_1	LED Link
GPIO[18]	Test Point	Test Point
GPIO[19]	Test Point	Test Point
GPIO[20]	Test Point	Test Point
GPIO[21]	Test Point	Test Point
GPIO[22]	Test Point	Test Point
GPIO[23]	Test Point	Test Point
GPIO[24]	NC	NC

Table 5-7. TILE-Gx16 or TILE-Gx9 Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[25]	NC	NC
GPIO[26]	NC	NC
GPIO[27]	NC	NC
GPIO[28]	NC	NC
GPIO[29]	NC	NC
GPIO[30]	NC	NC
GPIO[31]	NC	NC
GPIO[32]	NC	NC
GPIO[33]	NC	NC
GPIO[34]	NC	NC
GPIO[35]	NC	NC
GPIO[36]	NC	NC
GPIO[37]	NC	NC
GPIO[38]	NC	NC
GPIO[39]	Test Point	Test Point
GPIO[40]	NC	NC
GPIO[41]	Test Point	Test Point
GPIO[42]	NC	NC
GPIO[43]	NC	NC
GPIO[44]	NC	NC
GPIO[45]	NC	NC
GPIO[46]	NC	NC
GPIO[47]	Test Point	Test Point
GPIO[48]	XAUI0_LASI	XAUI PHY Link Alarm Status InterruptSGMII0_INT
GPIO[49]	XAUI0_LASI	XAUI PHY Link Alarm Status InterruptSGMII0_INT
GPIO[50]	XAUI0_LASI	XAUI PHY Link Alarm Status InterruptSGMII0_INT

Table 5-7. TILE-Gx16 or TILE-Gx9 Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[51]	XAUI0_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[52]	XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[53]	XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[54]	XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[55]	XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[56]	XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[57]	XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[58]	XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[59]	XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[60]	XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[61]	XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[62]	XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[63]	XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt

Note: There are three different I²C address that are active on a SPD ROM, which are referenced in [Table 5-8](#).

5.7.1 TILE-Gx Processor's I²C — Master Interface

The TILE-Gx Processor has three I²C Master interfaces and one I²C slave interface. In the TILEncore-Gx design, the slave interface is unused.

The three master and one slave interfaces are connected to the devices shown in [Table 5-8](#).

Table 5-8. I²C Address and Device Map

I ² C Address 7-Bit Address	Bus	Vendor and Part Number	Description
1010 100X	I2CM0	Microchip-24LC256-I/SN	IC. Serial EEPROM. 256K I ² C User Configuration Data
1010 000X	I2CM1		The SODIMM has an SPD ROM. There are three different I2C addresses that are active on a SPD ROM: • Serial Presence Detect • Temperature sensor • EEPROM Write Protection
0011 000X	NOT USED		
0110 000X	NOT USED		

Table 5-8. I²C Address and Device Map (continued)

I ² C Address 7-Bit Address	Bus	Vendor and Part Number	Description
1101 010X		IDT-IDT5v49EE904-032	Clock Driver Chip
0101 100X		Maxim-MAX6639/QSOP-16	Thermal Detect Chip

5.7.2 TILE-Gx Processor 4-Wire Interface

- TILE-Gx Processor SPI FLASH Memory

The SPI ROM on the TILE-Gx can be used for boot loading or other user configuration data. It is connected to the TILE-Gx's SPI interface. The device is a W25Q128FVSIQ, 128Mbit part.

5.7.3 TILE-Gx External Interrupts

Refer to [Table 5-9](#), which details the interrupts for the tile.

Table 5-9. TILE-Gx Processor's Interrupt Assignments

Interrupt Name	Interrupt Source	Physical Connection
POWER_INT_L	LC4064ZE CPLD	TILE-Gx: GPIO[43]
TEMP_INT_L	MAX6639 Temperature sensor. When asserted, LOE provides an early warning that power to the board is going down due to detection of a high temperature.	TILE-Gx: GPIO[44]
XAUI0_LASI	VSC8488	TILE-Gx: GPIO[48-51] ^a
XAUI1_LASI	VSC8488	TILE-Gx: GPIO[52-55]

a. Note that only two lanes are connected (XAUI0_LASI and XAUI1_LASI).

5.8 LEDs

5.8.1 Power and Status Indicators

Power and Status indicators, which include two Ethernet-status LEDs and two power-related LEDs, are listed on [Table 5-10](#). For information on each LED's behavior, refer to the *Descriptions* column.

[Figure 5-1](#) shows the location of the All Power Ready and GX Ready LEDs.

Table 5-10. Board LED Definitions

Reference	Color	Label	Location	Descriptions
D1	Green	GX Ready LED	Front	This LED indicates that the core RefClk is present, reset has been removed, and the internal configuration is complete.

Table 5-10. Board LED Definitions (continued)

Reference	Color	Label	Location	Descriptions
D2	Green	All Power Ready LED	Rear	This LED indicates that all power works on the card.
D3	Green	Ethernet Port 1 LED	Front	When the cable is plugged into the transceiver, the LED glows a steady GREEN. When the port is receiving data, the LED blinks GREEN, indicating there is activity on the port.
D4	Green	Ethernet Port 0 LED	Front	When the cable is plugged into the transceiver, the LED glows a steady GREEN. When the port is receiving data, the LED blinks GREEN, indicating there is activity on the port.

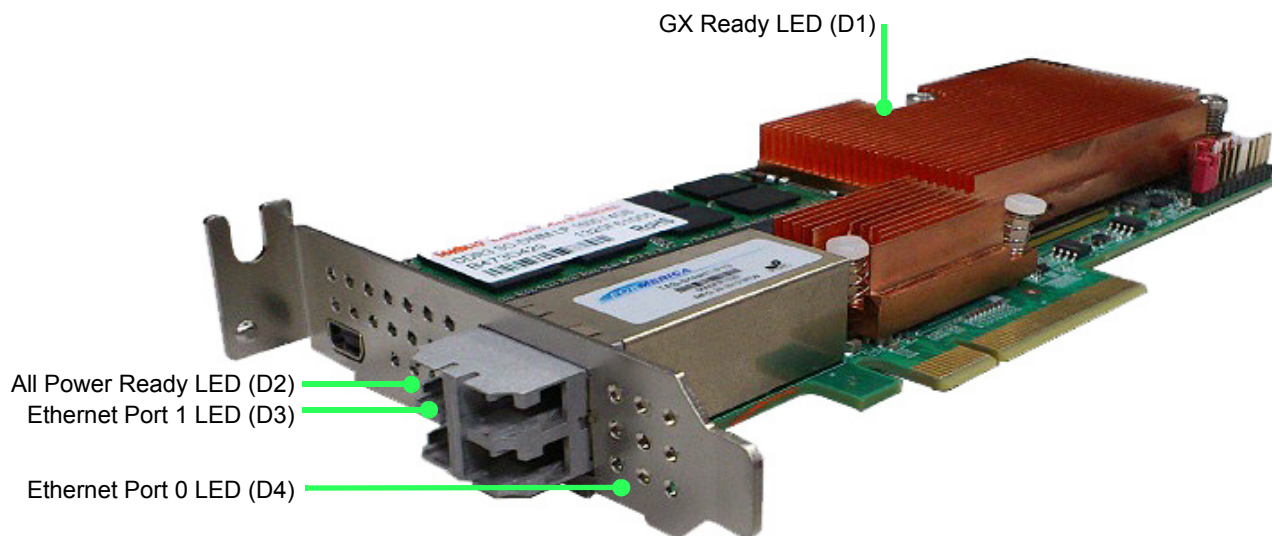


Figure 5-1: TILEncore-Gx Board LEDs — Top Side

5.8.2 Power LED

The power LED is located on the bracket (see [Figure 5-2](#)) and glows green only when all power components are power on.

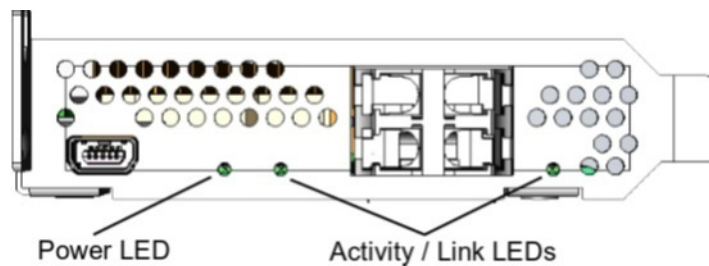


Figure 5-2: Link and Power LED

5.8.3 Port Link Status/Activity LEDs

The LINK LEDs, located on the bracket and depicted in [Figure 5-2](#), indicate that the XAUI links are operating properly.

CHAPTER 6 RUNNING PROGRAMS ON THE TILENCORE-GX16 OR TILENCORE-GX9

6.1 Considerations with Simultaneous USB and PCIe Usage

Booting the TILEncore-Gx™ with `tile-monitor` using the USB port is not supported, and can cause the PCIe host to crash. The board should always be booted over the PCIe port. You can, however, use `tile-monitor --resume` over the USB port to connect to an already booted board.

The TILEncore-Gx card supports boot and debug operations via its PCIe interface, as well as via its USB device interface. It is possible to connect both of these interfaces simultaneously, but doing so can cause undesired system behavior. This section explains how to avoid problems.

The problem is related to booting the TILE-Gx™ processor whenever you run `tile-monitor` *without* the `--resume` option, or whenever you run any chip tests via the Board Test Kit (BTK). The first step of booting the TILE-Gx processor involves resetting it. This operation is done automatically by the `tileusb` or `tilegxpci` drivers when you start to boot the processor via the corresponding interfaces.

When the processor is reset, all of the I/O interfaces on the chip get reset. In the case of USB, the USB device appears to be disconnected from the host; in the case of PCIe, the PCIe link between the host and the processor goes down. Soon afterward, communication is re-established.

In the case of USB, the link is brought back up automatically by the processor; in the case of PCIe, the PCIe prebooter is loaded from the SROM and brings up the link. However, the host's response to these momentary interruptions is very different. USB is designed as a hot-pluggable interface. The cabling for USB devices is generally accessible to their users, and many of those devices are connected to their host system only temporarily (as when you download photographs from a digital camera). Because of this, the USB software stack expects devices to be disconnected and reconnected.

While PCIe devices can also be hot-pluggable, this is the exception, rather than the rule. Desktop computers, which are commonly used to host TILEncore-Gx cards, typically do not support hot-plugging. Thus, their PCIe software stacks do not expect PCIe devices to be disconnected.

In order to cope with the fact that rebooting (and thus resetting) a TILE-Gx processor causes the USB or PCIe connection to be interrupted, the MDE's `tileusb` and `tilegxpci` drivers perform special steps as part of a reboot operation. The driver might need to read the configuration state from the device, though state information is destroyed by the reset. Once the device has been reconnected, the configuration state is restored. The driver might also need to reconfigure the system temporarily to tolerate the upcoming disconnect event. However, these steps can only be performed if the relevant driver knows that a reset is about to happen, which brings us to the problem with simultaneous USB and PCIe connections.

Imagine what happens when a TILEcore-Gx card is configured into a host system and both its USB and PCIe ports are connected. Both the `tileusb` and `tilegxpci` drivers believe they are responsible for controlling the card, and both devices are willing to reboot and thus reset it. However, neither driver has any “knowledge” of the other. Thus, if you use the `tileusb` driver to reset the card, the `tilegxpci` driver does not get the opportunity to do any of the normal state-saving and disconnection preparations; the host system sees an unexpected interruption of service to one of its PCIe devices instead.

The effect of this kind of event is system-dependent. On some systems, the unexpected PCIe disconnect leads to a fatal error and a system crash. On others, the system does not crash, but the fact that the newly-reset TILEcore card has lost the configuration information programmed into it during device discovery can lead to problems with PCIe I/O, perhaps not even involving the TILEcore card. In either case, the only way to recover is to reboot the host system.

Because of this, we strongly recommend that if you have a TILEcore Card connected via both the PCIe and USB, that you never reboot the card via USB.

You should not boot via USB. Specifically, you should not run `tile-monitor` against the USB interface, and should not run any tests when using `tile-btk` against the USB interface. Rebooting such a doubly-connected card via PCIe, on the other hand, is generally safe, since the USB interface is much more tolerant of momentary interruptions.



If you have PCIe and USB connected at the same time, **you should avoid resetting the card via USB.**

If you reset the card over USB, expect one of two things is going to happen:

- If your card is in a host **that is not sensitive to the PCI link going down**, the PCI link goes down and comes back up, but the chip **is not** programmed properly and you are be able to communicate with the card over PCIe. It is also possible that the default settings on the card might interfere with other things on the BUS, which is less likely but still possible. The main symptom is that you are not able to talk to the card over PCIe. The only way to fix that problem is to reboot the host system.
- If your card is in a host **that is sensitive to the link going down**, the system most likely crashes and reboots. Some hosts might not be set up to reboot after a failure like that. You might need to power cycle the host in these cases.

6.2 Boot Options

The TILEcore-Gx card supports the following boot options:

- Boot from host via the TILEcore-Gx™ card’s PCIe 0 or 1 Interface. This is the standard boot option.
- Boot from boot ROM (SPI ROM). This option requires programming of SPI ROM.

Note: If the board is booted from SPI-ROM, it will not be able to communicate with the host over PCIe.

You can also run against an already-booted TILEcore-Gx card without rebooting it by using the `--resume` option:

```
tile-monitor --dev usb0 --resume
```


6.2.1 Booting a TILEncore-Gx via a PCI Interface

```
tile-monitor --dev gxpci0 --hvx TLR_NETWORK=none
```

For more information about tile-monitor options, refer to *Getting Started with the Multicore Development Environment* (UG504).

```
tile-monitor --dev gxpci0 --resume
```

6.2.1.1 PCIe-Related tile-monitor Arguments

`--pci-link number`

Use the `--pci-link` argument to specify which PCIe link to use (defaults to 0). Using this argument might be useful when running multiple clients on a single PCIe card, with a separate shepherd running in each client.

`--pci-stream number`

Use the `--pci-stream` argument to specify which PCIe stream to use to communicate with the shepherd (defaults to 0).

APPENDIX A: SPECIFICATIONS

A.1 Electrical Specifications

Power consumption varies depending on the user's application, but [Table A-1](#) can be used as a guide:

Table A-1. Power Consumption for TILEncore-Gx Card

Card Version	Typical	Maximum
TILEncore-Gx36 v2 card @ 1.0 GHz	25W (1.6A @ 12V & 1.4A @ 5V)	30W

See also [Section 3.2 "Power Requirements"](#) on page 12.

Note: The board consumes no more than 117 watts.

A.2 Mechanical Specifications

The mechanical dimensions of the card are compliant with the add-in card form factor for an 8-lane PCI Express "short" add-in card.

Table A-2. Physical Specifications for the TILEncore-Gx Card

Board Height	64.4 mm (2.54 inch)
Board Length	168.15 mm (6.6 inch)
Board Depth	18.54 mm (0.73 inch)

A.3 Environmental Specifications

A.3.1 Environment

Table A-3 lists the other environmental specifications of the TILEncore-Gx card.

Table A-3. Chassis-Level Environmental Specifications

Specification	Range
Storage temperature	0° C to +40° C This represents ambient temperatures surrounding the TILEncore-Gx card.
Operating temperature	0° to 45° C
Operating humidity	5% to 95%, non-condensing
Vibration	0.5G (Peak) / 5 500 Hz, 2hours at each of Z axis.

A.3.2 RoHS Statement

The TILEncore-Gx is built to comply with the 2002/95/EC European legislation of 2003-02-13, which states that equipment must be free of:

- Lead
- Mercury
- Cadmium
- Hexavalent Chromium
- Polybrominated Biphenyls (PBB)
- Polybrominated Diphenyl Ether (PBDE)

Products free of the materials is known as being RoHS-6 compliant. However, the TILE-Gx process has a small amount of lead contained in the internal die bond for the “flip chip” package. This means that at the board level, this product is specified to be RoHS-5 compliant. However, there are no other devices containing lead on the PCB, and the soldering technology for all components is lead-free.

APPENDIX B: SAFETY AND REGULATORY COMPLIANCE

B.1 Introduction

This chapter explains important safety precautions that you should observe when operating the TILEncore-Gx™ card.

Note: Follow all warnings and instructions marked on the equipment.

B.2 Safety Precautions

B.2.1 Use Proper Grounding

To avoid electric shock, make sure that the system is connected to an electrical safety (green wire) ground.

B.2.2 Use ESD Protections

The TILEncore-Gx contains static-sensitive devices. Static control procedures should be used when installing, removing, and handling all components. Use an antistatic wrist strap at all times. Wrist straps usually come with an alligator clip that you clip to the metal frame of the chassis after the cover is removed.

Never touch the Card with any conductive objects.

For additional handling instructions, refer to [Section 4.2.2 “Handling the TILEncore-Gx Card”](#) on page 13.

B.2.3 Avoid Fumes and Explosives

Do not operate the TILEncore-Gx card in the presence of explosives or inflammable gases or fumes.

B.3 Regulatory Compliance

B.3.1 Emissions and Susceptibility Certifications

CE/FCC class A for server/equipment.

B.3.2 FCC Notice

This equipment generates, uses, and can radiate radio frequency energy. If the Card is not installed correctly and used in accordance with the instruction manual, it can cause interference to radio communications.

APPENDIX C: TROUBLESHOOTING

This chapter describes basic methods for troubleshooting your TILEncore-Gx™ card.

How do I reinstall the PCIe prebooter?

Boot the board using USB as described in [Section 6.2 “Boot Options”](#) on page 34. Connect a serial console device to the UART.

From the console port, type:

```
sbim --info
sbim --install-prebooter
```

First you have to upload the pre-booter from `tile-monitor` as follows:

```
tile-monitor --dev gxpci0 --resume --upload-tile /boot/pci-preboot.bin
```

At the prompt, respond with “Yes” and then enter:

```
sbim --info
```

The system responds with something like:

```
SROM booter: Present 8336 bytes
Time Installed: Tue
```

What does tile-console do?

The `tile-console` command provides access to the console port on a Tile Processor, which displays Linux boot messages and normally provides an interactive Linux login, via a serial device on the host.

Use the same device name nomenclature as with `tile-monitor`. For example:

```
--dev usb0
—OR—
--dev gxpci0 (default)
```

With tile-btk operations, I cannot reboot the TILEncore-Gx card with --dev gxpci0 after boot it with --dev usb0. Is this expected?

Yes, you can expect this behavior. We recommend power-cycling the host *after* every time you run the btk commands.

Even if the btk can reboot the TILE-Gx from the SROM and potentially get the PCIe link up, the host does not assign any BAR space to the TILE-Gx, unless we support PCI re-scan, which we do not. Therefore, no further PCI communication can happen.

APPENDIX D: GLOSSARY

Term	Definition
2-WIRE	2-Wire Serial Clock and Data (SCL and SDA) operating up to 400kHz.
4-WIRE	4-WIRE, or two pair, Synchronous Serial Interface or Serial Peripheral Interface (SPI).
BIB	Board Information Block. The Tiler [®] run-time software stack uses a set of system-specific information to govern the configuration and initialization of the Tile Processor [™] and the devices to which it connects. This data is contained in a structure called the Board Information Block (BIB).
CPLD	Complex Programmable Logic Device
DDR3	Double Data Rate, Type 3. Synchronous dynamic random access memory—a modern kind of dynamic random access memory (DRAM) with a high bandwidth interface.
DIMM	Dual In-line Memory Module • LRDIMM – Load reduced • RDIMM – Registered • UDIMM - Unbuffered
ECC SODIMM	Error Check and Correction Small Outline DIMM.
EEPROM	Electrically Erasable Programmable Read-Only Memory.
GbE	Gigabit Ethernet.
GPIO	General Purpose I/O.
I/O	Input/Output.
LED	Light-Emitting-Diode used for status indicator
LVPECL	Low-Voltage Positive-referenced Emitter-Coupled Logic
MDE [™]	Multicore Development Environment [™] . Tiler's multicore programming environment.
PCI	Peripheral Component Interconnect, a bus standard for connecting peripheral devices to a computer motherboard.
PCIe or PCI-e	PCI Express, a PCI interface format using high-speed serial links for the connection between peripheral devices and the motherboard.
ROM	Read-Only Memory.
SFF	Small Form Factor.
SODIMM	Small Outline DIMM.
SPD ROM	Serial Presence Detect ROM.

Term	Definition
SPI	Serial Peripheral Interface.
SPI-SROM	Serial Flash with serial peripheral interface.
UART	(Universal Asynchronous Receiver Transmitter). The electronic circuit that makes up the serial port. Also known as "universal serial asynchronous receiver transmitter" (USART), it converts parallel bytes from the CPU into serial bits for transmission, and vice versa. It generates and strips the start and stop bits appended to each character.
USB	Universal Serial Bus
VID	Voltage Identification (bits)
XAUI	10 Gigabit Attachment Unit Interface, a standard for connecting 10 Gigabit Ethernet.

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